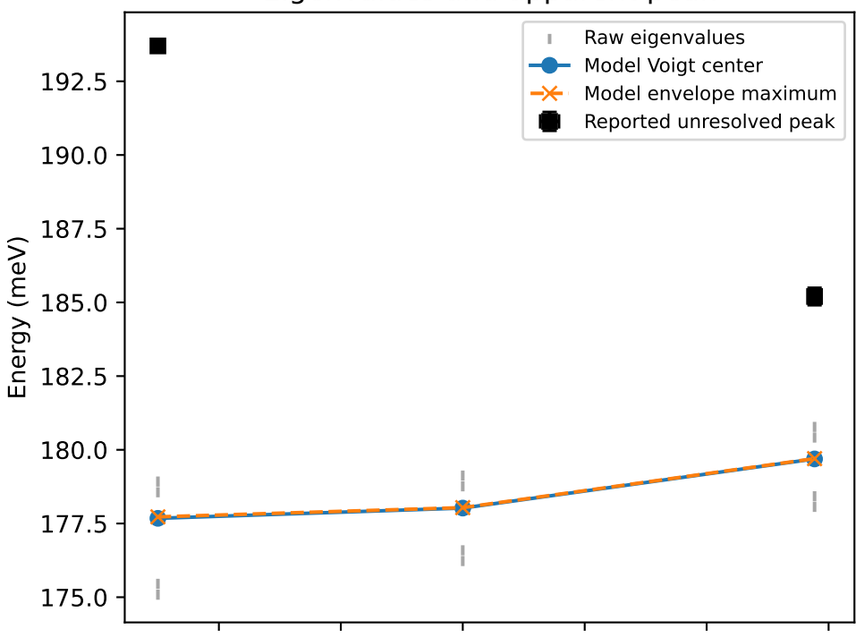
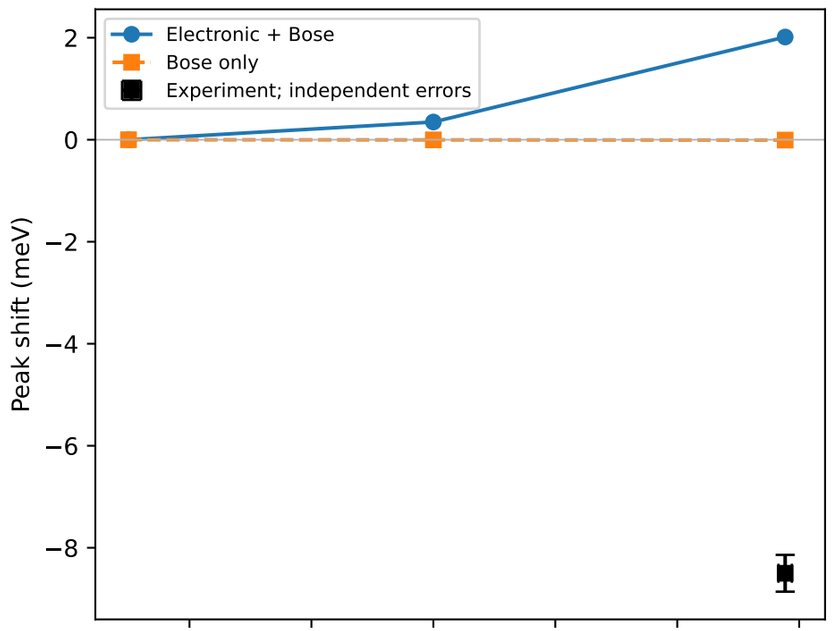
AB graphite PBE baseline, pilot: unconverged basis/k/FFT; fixed geometry
Unresolved-peak comparison; no empirical frequency correction

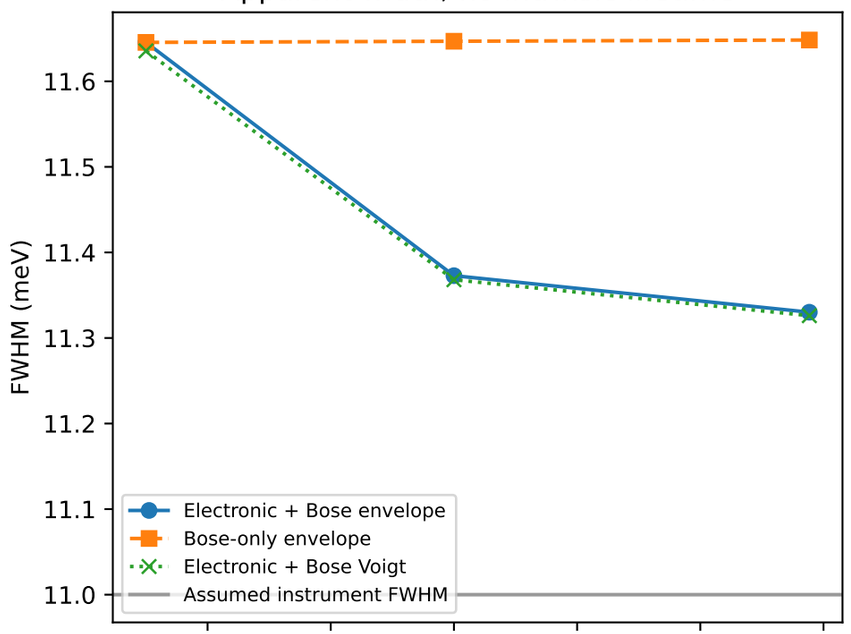
Eigenvalues and apparent peak



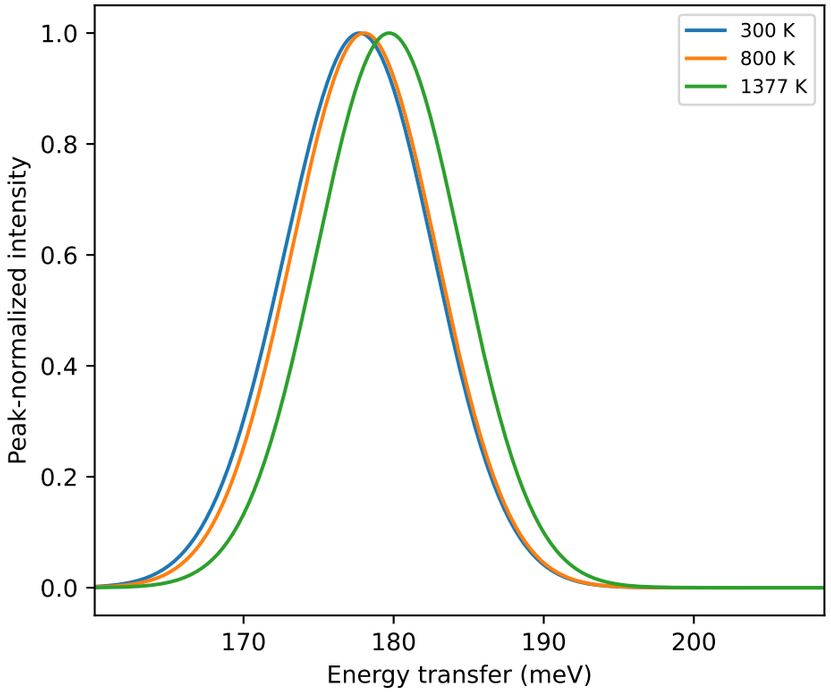
Shift relative to 300 K



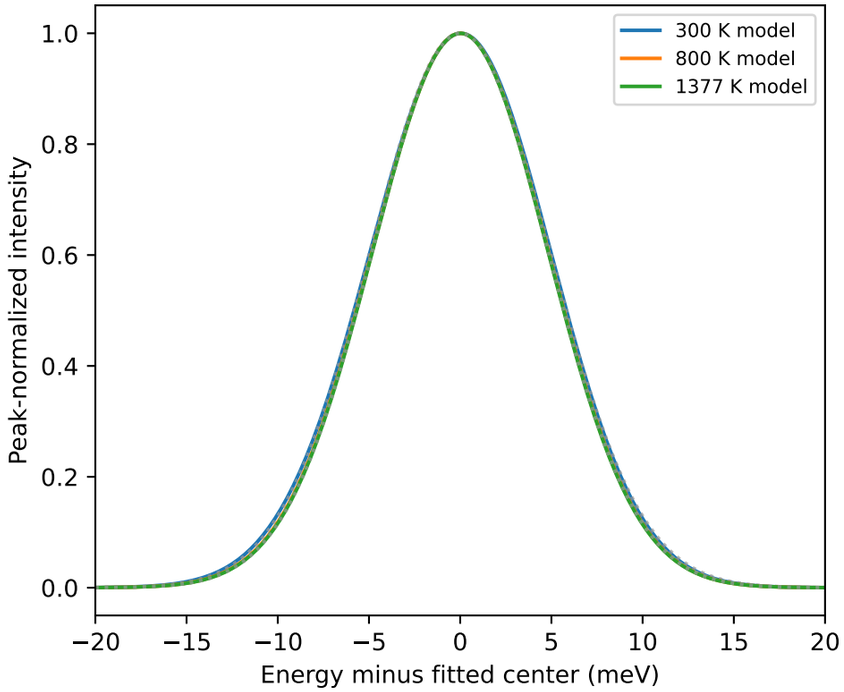
Apparent width; no intrinsic lifetime



Normalized model loss shapes



Centered shapes; dotted = Voigt fits



Comparison boundaries

Experimental static shift: -8.5 meV
Formal error: 0.361 meV, independent endpoints.
Room temperature mapped to 300 K.

Reported ~7 meV broadening concerns Fig. 4f time-resolved data, not the static endpoints.
No exact width endpoints or errors supplied.

Model: one Q, unit form factors, Gaussian 11 meV.
No momentum-window integration.
No thermal expansion or bulk anharmonicity.
No model uncertainty or lifetime prediction.

Source: arXiv:2610.18561v1, II / III.1 / IV.